Intelligent Automated Taxis

PEAS Analysis

PEAS description of the task environment for an automated taxi involves evaluating the agents involved in the environment based on the following four components:

1. **Performance Measure**

The taxis can navigate and operate safely and get to the destination fast. The performance measure can be defined as the number of successful trips completed per hour, the average passenger rating received, or the total revenue generated.

2. **Environment**

It includes factors such as the traffic conditions, weather conditions, road infrastructure, availability of parking spaces, and the behavior of other vehicles and pedestrians on the road.

3. **Actuators**

In the context of an automated taxi, the actuators could include the steering system, accelerator and brakes, speakers and display screens.

4. **Sensors**

Sensors provide information to the automated taxi about the current state of the environment and its own internal state. In the case of an automated taxi, sensors could include cameras, LIDAR, ultrasonic sensors, and radar for detecting and perceiving the surrounding vehicles, pedestrians, and road conditions. Additionally, the taxi may have sensors to monitor its own internal systems, such as battery charge and engine health.

Task Environments for an Automated Taxi

- Fully vs. **Partially Observable** Environments
  - Partially observable environments result from noisy sensors or missing data, requiring agents to make decisions based on incomplete information.
  - Obstacles can block the vision sensors.

- Single Agent vs. **Multiagent** Environments
  - Multiagent environments feature interactions between multiple taxis, influencing each other's performance measures.
  - Competitive multiagent environments involve taxis maximizing performance measures that conflict with each other.

- Episodic vs. **Sequential** Environments
  - taxi driving exemplify sequential environments where short-term actions impact long-term results.

- Static vs. **Dynamic** Environments
  - Taxi driving exemplifies a dynamic environment with continuous movement.

- **Deterministic** vs. Stochastic Environments
  - Taxis can always reach the destination.

- Discrete vs. **Continuous** Environments
  - Taxi driving involves continuous states, time, and actions.
  - Continuous environments involve smooth transitions over time.
  - Taxi driving actions are continuous (e.g., steering angles).

- Known vs. **Unknown** Environments
  - Unknown environments necessitate learning the environment's dynamics of customer behavior.

Agent Architecture

(Utility-Goal-and-Model-based Architecture)